

During the COVID-19 pandemic, hand washing has become a no-brainer, an intuitive must-do to rid your hands of bacteria and contamination. Hand washing is a preventative measure for contaminants that can lead to diarrhoeal disease, acute respiratory infections, and influenza¹. Infection control is a staple in the hospital where diseases such as bacterial meningitis, measles, West Nile virus, and hepatitis A, among others, can be contracted². Public health experts have stated that although hand washing consistently halts the transmission of most infections, compliance with such standards has been one of the greatest challenges in hospitals. In a teaching hospital in Geneva, Switzerland, the average compliance with handwashing procedures was 48%³. Suppose the “father of infection control” was aware of the present-day hand-washing practices. In that case, it must be a bittersweet feeling as although he has received recognition and his procedures are widely advocated for, his recognition was much overdue and his findings are often misunderstood.

Ignaz Semmelweis was a Hungarian physician and scientist who was born in 1818 and raised in Austria. Although Semmelweis’ legacy and main focus was medicine, he had always appeared to be a very interdisciplinary scholar. Initially drawn to law during his collegiate studies, Semmelweis’ curiosity soon veered toward medicine in 1837 during his time at the University of Vienna. Later on, Semmelweis also delved into the field of English and was fascinated by art.

Following his graduation and the reception of his Doctor of Medicine degree, Semmelweis started work at the first division of obstetrics clinic of the Vienna General Hospital in 1846. During this time, the mysterious postpartum infection, also known as puerperal fever or childbed fever, was high and feared by many. As Semmelweis recounts in the autobiographical introduction of his book, “The Etiology, Concept, and Prophylaxis of Childbed Fever,” at the Vienna General Hospital, there were two divisions of obstetrics clinics where all male students — medical students — and physicians were assigned to the first clinic and all female students — midwife trainees — and midwives were assigned to the second clinic⁴. The two divisions were identical in every aspect, from medical procedures to facilities, except for the gender separation of the staff which was due to social norms and gender roles at the time. Over time, Semmelweis noticed that the mortality rate in the first clinic was consistently higher than the mortality rate of the second clinic, at 13-18% versus 2%, respectively⁵. This alarming contrast prompted him to contemplate the reason for the vast difference in mortality rates between the two obstetric clinics where risks such as hyperinosis (abnormal excess of fibrin in the blood), weak blood circulation, imperfect contractions, and delivery itself, in addition to other dangers, all pose the same risk regardless of the clinic and who delivered. Both clinics used the same techniques, the same equipment, and had the same resources. Thus, Semmelweis started to consider unexplainable factors, such as the occurrence of childbed fever in both the mothers and newborns.

As depicted in Semmelweis’ medical book, the mortality rates from 1841-1846 for all patients in the first and second clinics were 9.92% and 3.38%, respectively. The mortality rates for newborns in the first and second clinics during the same time frame were 7.55% and 3.63%, respectively⁴. Since there was the same trend of mortality rates in both mothers and newborns at

both clinics, Semmelweis could not attribute the source of the mortality rates to be due to the anatomy of the adult versus the child.

He observed that not only can childbed fever be contracted during or after birth, but Semmelweis noticed that even mothers who gave street births contracted the fever at a significantly lower rate than those who delivered in the first clinic. Street births are births where infants are born “on the street,” or on the way to the clinic, in order for the women to receive the same childcare benefits as if they gave birth in the clinic. It would be intuitively assumed that these births would pose a greater risk of infection or contamination of childbed fever due to the lack of hospital facilities and staff, but this was not the case as the mortality rate was comparable to that of the second clinic. Eventually, Semmelweis determined that the cause of the high mortality rate due to infection of childbed fever was limited to the first clinic.

In 1847, Semmelweis was still grappling to understand the reason behind the gap in mortality rates between the first and second clinics. It was when he heard of Professor of Forensic Medicine Jakob Kolletschka’s passing that he started to put the pieces together. Kolletschka was conducting an autopsy with a group of students when his finger was pricked by the same blade used in the autopsy. A few days later, the professor passed away with the same symptoms that claimed the lives of the women in the first clinic. This was the moment Semmelweis had his epiphany and concluded that the higher mortality rate of the first clinic was due to the women’s exposure to cadavers prior to delivery, which led to childbed fever being induced. While physicians and medical students in the first clinic had the opportunity to handle corpses during autopsies, midwives and midwife trainees in the second clinic didn’t.

Semmelweis believed contact with cadavers to be the source of childbed fever, building upon his knowledge that when decomposing organic material is brought into contact with living organisms, decomposition may occur. Thus, Semmelweis implemented the unique, rarely executed practice of hand washing in May 1847 in an attempt to eradicate cadaverous bacteria from the hands. In the first clinic, he required a hand washing policy for all medical students and physicians to wash their hands using chlorinated lime, more commonly known as calcium hypochlorite, between autopsies and patient interactions. Calcium hypochlorite is primarily used as a bleaching agent or disinfectant for drinking water, as found on the Centers for Disease Control and Prevention website⁶. Numerous experiments have found that when calcium hypochlorite is dissolved in water, the disinfectant destroys a significant amount of bacteria⁷.

So, Semmelweis hypothesized that the cadaverous bacteria on the physicians’ and medical students’ hands could be eliminated chemically through hand washing with water that has gone through the hypochlorite process, therefore decreasing the onset of childbed fever. The effectiveness of these new procedures was observed immediately: from June to December of 1847, the mortality rate of all patients in the first clinic decreased to 3.04%. In March and August of 1848, not a single patient died in his clinic⁴. Semmelweis, the “savior of mothers,” had just discovered that infection could be reduced by hand washing — the start of antiseptic procedures.

Unfortunately for Semmelweis, the germ theory was not yet established and popularized, and it would be until the end of the 19th century that it was. German physician and

microbiologist Robert Koch established the germ theory almost four decades later in 1884, which stated that microscopic germs, or pathogens, cause disease⁸. Koch formed his theory using four postulates, now known as Koch's postulates, during his experimentation with anthrax bacteria. The four postulates are: 1) The microbe is always present in the diseased host and absent in healthy individuals; 2) The microbe must be isolated from diseased individuals; 3) The healthy individual becomes ill when the pure microbe is introduced into the host; 4) The same microbe can be re-isolated from the inoculated individual. It would happen so that Semmelweis' findings of childbed fever being an infected wound disease would support the development of the germ theory⁹.

Before the germ theory, the miasma theory was the widely accepted and dominating theory of disease transmission. Late University of Colorado, Boulder psychology and neuroscience Professor Emeritus Bernard Bloom explained that the miasma theory "held that soil polluted with waste products of any kind gave off a 'miasma' into the air, which caused many major infectious diseases¹⁰." This miasma, or "bad air" from rotting organic matter, was the explanation for diseases such as cholera and the Bubonic Plague¹¹. So, following the release of his book in 1861 where he documented his findings of the cause and preventive measures of childbed fever, the scientific and medical community rejected him as there was no theoretical backing for Semmelweis' medical breakthrough.

Having made the novel discovery in his late 20s, Semmelweis' young age caused the scientific community to perceive him as inexperienced, especially as his ideas challenged the orthodox medical doctrines and authority of well-established seniors. This, coupled with the proposal that such a simple explanation of cleanliness and handwashing could drastically reduce the enigma that was childbed fever, had other doctors and scientists ridiculing him. Even more so, many of the scientific community were angered and offended by the seemingly simple suggestion as they felt that Semmelweis was accusing them — educated, clean people — of being unclean. This led to Semmelweis being ostracized.

However, Semmelweis stood his ground unyielding against those who dismissed him. Although "uprightness, respectability, goodnaturedness are the principle features of his character," as described by Semmelweis' friend and colleague János Bókai, he was also "very gruff and merciless in his utterances, he whom he dealt with rudely in passion¹²." This duality became evident in his open letters that were a response to those who dismissed the findings, where Semmelweis addressed doctors, professors, and scientists in an increasingly angry, bitter, and offensive manner. "You, Herr Professor, have been a partner in this massacre" and "I declare before God and the world that you are a murderer," are among some of the statements he made to the scientific and medical community⁵. The Hungarian doctor's frustrations and ego became the pitfall in not only his journey to gain career recognition and respect but also his life.

Prior to the publication of his book, the public perception of Semmelweis' character was already skewed as his political associations with the Hungarian independence movement in 1848 caused uneasiness in those around him¹³. Although there was no confirmation that Semmelweis participated in the movement, his brothers did partake. The suspicion surrounding his personal

life led the medical community to distrust his nature, and also question the dependability of his professional work. This amplified the obstacles Semmelweis was facing in pushing for handwashing procedures even before the publication of his book.

The scorn and incredulity of the scientific community took a significant toll on his mental health, overshadowing his potential in the medical field. Semmelweis became increasingly erratic and suffered from severe depression, leading his colleagues and wife to believe that he was losing his mind. Bókai and other colleagues noticed his behavior that became increasingly uncharacteristic: He was indulging in alcohol in an unhealthy manner, had heightened sexual excitement, had a relationship with a prostitute (a stark contrast to his previous unwavering devotion to his wife), was wasteful with finances, and had indecent behavior in interpersonal relationships, among others. In 1865, Semmelweis was manipulated by his peers. Under the facade that he was visiting an asylum that was newly opened by his colleagues, Semmelweis sensed too late that he was the one to be admitted into the asylum. Upon realizing this betrayal, he was beaten by guards and bound in a straightjacket when he tried to escape. He ultimately succumbed to a wound infection on his right hand — ironically, the field that he devoted much of his life to investigating became his undoing.

Nevertheless, Semmelweis' insights should be clarified to prevent any misinterpretation that can lead to simplifying his idea. Many were unreceptive to his ideas as they were too easy to assume that Semmelweis was advocating for normal hand washing with soap when, more specifically, he was advocating for disinfection — a concept that was advanced for his time and thus met with skepticism from the scientific community. Today, the low compliance with recommended handwashing practices of 10-50% in hospitals is not due to a lack of cleanliness, but can rather be attributed to the preference for alcohol-based disinfectants — what Semmelweis was championing¹⁴. A study published in the peer-reviewed medical journal *Clinical Infection Diseases* explains how Semmelweis didn't advocate for handwashing with soap and water, but more specifically with chlorine which is extensively used as a disinfectant and antiseptics¹⁵. The *Journal of Hygiene* has additionally found that although there is a fairly good correlation of reduction in bacteria between hospital nurses who frequently complied with hand washing procedures versus volunteers who didn't, hand washing with soap only did not remove all patient-borne bacteria¹⁶. Alcohol-based solutions or antiseptics such as 4% chlorhexidine detergent scrub or iodophor were found to be effective in removing tough bacteria. These alternatives produce quicker hand hygiene with a higher rate of bacteria elimination. Semmelweis' original hypothesis was more specifically about hand disinfection, which can be achieved to a certain extent through hand washing.

This event truly reveals the cultural and social influences on scientific investigation, supporting the radical subjectivist model more than Popperian ideals, although both can be explained by this historical event. Whilst Semmelweis' hand-washing theories were testable and falsifiable, aligning with Popperian ideals, social expectations and stubborn pride in the medical community prevented the theory from taking off. Additionally, Semmelweis addressed the specific problem of high maternal mortality rates through a problem-solving approach, in line

with Popperian ideals. However, since his theories had no evidence backing them at the time, they defied the Popperian ideal of refutability. The radical subjectivist model takes a more creative, open-minded approach influenced by external contexts, such as scientific consensus as a social construct — which is the foundation of the tragic history of Semmelweis. Semmelweis' work is not only a scientific milestone but a formidable example of the generally traditionalist scientific community. At the time, the miasma theory was already widely accepted, leaving no room for a new theory and highlighting the resistance to new, unconventional ideas. The refusal to accept Semmelweis served as the drive in the evolution of the germ theory, laying the foundation for microbiology, in addition to illustrating the multifaceted nature of scientific investigation.

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